

Confidence-Based Dynamic Agent Orchestration for Adaptive Multi-Agent Retrieval-Augmented Generation

Introduction

Retrieval-Augmented Generation (RAG) enhances Large Language Models (LLMs) by retrieving relevant information from external knowledge sources before generating responses, reducing hallucinations and improving factual accuracy. Recent Multi-Agent RAG systems further improve performance by employing specialized agents for retrieval, reasoning, verification, and response generation. However, most existing frameworks follow a fixed execution workflow in which every agent is invoked for every query, regardless of its complexity or the reliability of intermediate results. This leads to unnecessary computational overhead, increased response latency, and inefficient resource utilization.

Problem Statement

Current Multi-Agent RAG systems lack a confidence-driven decision mechanism to determine whether additional retrieval, reasoning, or verification is actually required. As a result, computationally expensive agents are executed even for simple queries, while the system cannot intelligently adapt its workflow based on the reliability of its intermediate outputs.

Proposed Solution

This research proposes a **Confidence-Based Dynamic Agent Orchestration** framework for Adaptive Multi-Agent RAG. Instead of executing all agents sequentially, the proposed framework estimates the confidence of intermediate outputs and dynamically decides whether to return the response, invoke a verification agent, perform additional retrieval, or rewrite the query. The confidence threshold will be determined experimentally using benchmark datasets by evaluating multiple threshold values and selecting the one that provides the best balance between answer quality and computational efficiency.

Objectives

- Develop a confidence estimation mechanism for Multi-Agent RAG.
- Design a confidence-driven dynamic orchestration framework.
- Experimentally determine optimal confidence thresholds.
- Reduce unnecessary agent execution while maintaining response quality.
- Compare the proposed approach with conventional fixed-workflow Multi-Agent RAG systems.